

LED Dice Adventure Random Number Project

Simplified Curriculum & Lesson Guide

A fun, hands-on introduction to loop circuits, simple inputs, and electronic outcomes designed perfectly for early childhood classrooms.

Target Group:	Grade 1 (Ages 6-7)
Duration:	30 Minutes
Core Topic:	Simple Loop Circuits, Inputs, & Outputs
Architecture:	12-Page Simplified Blueprint

1. The Fun Story to Start the Class

| The Mystery of the Missing Dice

When teaching six-year-olds, starting with boring textbook terms like "current flow" or "resistance" does not work. Kids need a story they understand to make them excited to build things with their own hands.

This lesson starts with a story that almost every child in an Indian home knows: playing a fun board game on a rainy day, only for the game to stop because a tiny piece goes missing!



TELL THIS STORY TO THE KIDS:

"Imagine you are sitting happily on the big floor mat on a rainy Sunday evening. You are playing a fun game of Ludo with all your cousins, your big brother, and your sweet grandparents. The bright board is laid out right in the middle of everyone. You picked the blue token, and you only need three more steps to reach home and win the whole game!"

You shake your hands happily, ready to roll the dice, but oh no! The tiny white dice slips right out of your fingers, tumbles off the sheet, and rolls deep under the heavy wooden storage diwan couch!

It is too dark down there to see anything, and nobody can crawl underneath to get it. Does our game have to stop forever? Are we stuck? No way! We are smart little engineers. Today, we are going to build our very own magical, glowing, electronic LED Dice from scratch!"

Teacher Tip:

Use animated hand gestures when pretending to drop the dice under the diwan. Ask the children if they have ever lost a toy under a heavy bed or couch to make them feel part of the story.

2. What is a "Surprise Number"?

Explaining Random Numbers to 6-Year-Olds

The mathematical idea of "randomization" can sound scary to young children. Instead of using math definitions, we explain it as a fun, magical surprise!

Children learn that a random number is a secret surprise that the computer or the tool chooses entirely on its own. No player can guess what number is coming up next until it actually appears!

The Secret Paper Slip Game (Chit-Picking)

To help kids understand how a tiny computer chip switches through numbers quickly to choose one, the teacher can compare it to a popular game kids play every day:



EXPLAIN USING THIS SIMPLE EXAMPLE:

"Think about what happens when you play the paper chit game at school with your friends. You take a clean piece of paper, tear it into five small square pieces, and write a secret number or a friend's name on each one.

Next, you roll and fold every little paper slip into a tiny, tight ball. Now, they all look exactly the same from the outside! You put all the paper balls into Grandfather's old cotton cap or a shiny kitchen steel bowl. You mix them around and around, close your eyes tight, and pull one out.

Until your fingers open up that paper ball, the number inside is a total secret! That fun surprise is exactly how our electronic dice picks a number for your game."

Teacher Tip:

Keep a real steel bowl and a few folded paper slips on your desk. Quickly show them this mixing action so they can see and feel the idea of a "surprise choice".

3. Meet Our Circuit Friends!

Introducing Hardware Components as Characters

Instead of memorizing tough words like "positive anodes," "voltage," or "insulators," this lesson introduces hardware parts as friendly little characters who work together in a team. Each part has a special job to do to make the dice light up!

The Energy Tank (The Coin-Cell Battery)

This is a small, round, silver battery. Think of it like a mini water tank sitting high up on a roof. It holds all the electrical power inside, ready to rush down the wires whenever we open up a clear path for it.

The Gatekeeper (The Push-Button Switch)

This is a small button with a tiny spring inside. It acts just like the main door of your house or a water tap.

- When nobody is touching it, the door is closed and locked tight. The power cannot pass through.
- When your little fingers press down on it, the door swings wide open, and the power flows through easily!

The Beautiful Light (The LED Light Matrix)

This is the final stop for the electricity. When the power flows into this light grid, it instantly turns into a bright, glowing pattern of dots. This pattern shows us the surprise number we just rolled!

Language Reminder:

Always use these character names (Energy Tank, Gatekeeper, and Beautiful Light) during the class. It helps the children remember what each part does without getting confused by technical words.

4. Action and Reaction

Understanding Inputs and Outputs

An important skill for young engineers is understanding how actions cause results. We teach children that electronic gadgets work by linking a human action (the **Input**) to a machine reaction (the **Output**).

The Neighbor's Doorbell Example

USE THIS EVERYDAY EXAMPLE:

*"Think about walking up to your aunt's house or your neighbor's front door. You reach out your thumb and press down hard on the little plastic button on the wall. Your push is the **Input**!*

*Right away, from deep inside the house, you hear a loud 'Ding-Dong!' sound. That sound is the house's **Output**!*

*If you do not push the button, the doorbell stays completely quiet. In our electronic dice game, pushing the button is our **Input**, and the sudden flash of bright, glowing dots is our beautiful **Output**!"*

The Classroom Action Map

What You Do (Input Action)	What the Machine Does (Output Result)
You press down on the dice's push button.	The path links together and sends power rushing into the brain chip.
The inner brain chip picks a surprise paper slip.	Special LED lights instantly turn on to show a happy cluster of dots.
You lift your finger away from the button.	The path opens up, the power stops, and the lights go to sleep safely.

5. The Magic Circle

| The Idea of an Unbroken Circuit Loop

Electricity cannot jump across empty spaces in the air. It behaves exactly like a little flowing stream of water that needs a perfect, completely connected loop to move forward. If there is a tear, a break, or a loose fit anywhere along the path, the power stops moving right away, and the lights stay dark.

| The Temple Walking Path (Pradakshina)

To make this continuous circle clear to young children, we explain it using a path they walk with their families during community or temple visits:



EXPLAIN THE CIRCLE LIKE THIS:

*"When you visit a beautiful temple or a big community playground with your parents, you walk along the special stone path that goes all the way around the central building. This round walk is called a **Pradakshina** or **Parikrama**.*

To complete this special walk, you must start at the front gate, walk smoothly all the way around the back wall, and come right back to the front steps without missing any stones.

What happens if a giant metal fence blocks your path halfway through? Your circle breaks, and you cannot finish your walk! Electricity moves the exact same way. It needs a complete, unbroken walking path to run its course."

| Connecting the Path on the Board

When building the dice, the teacher will show that the power moves out from the positive side of the battery, travels through the button switch, enters the light frame, and goes right back down into the negative side of the battery to complete the circle.

6. The 30-Minute Adventure Map

Class Pacing for Teachers

Keeping the class moving at a fun, steady speed ensures that Grade 1 students finish building their electronic dice before they get tired or lose interest.

Kits must be set up neatly beforehand so the lesson fits perfectly into this 30-minute schedule:

Time Block	Activity Name	What to Do & Say in the Classroom
00:00 - 05:00 (5 mins)	The Story Hook	Tell the story of the missing dice under the heavy diwan. Ask kids what games they love to play at home (like Ludo, Snakes & Ladders, or Paramapadham).
05:00 - 12:00 (7 mins)	Meet the Parts	Open the kit boxes. Match the battery, button, and lights to the water tank, gatekeeper, and surprise slip game ideas.
12:00 - 25:00 (13 mins)	Guided Build	Follow the easy steps together. Click the power lines, fit the parts into the plastic shell, and test the connections.
25:00 - 30:00 (5 mins)	Ludo Fun Race	Let students use their new glowing dice to play a quick, 5-step mini game with their desk partners to make sure everything works perfectly.

7. How to Build: Step 1

| Setting Up the Power Without Any Worry

Grown-up electronics tools can sometimes be scary for small children because of sharp edges, scissors, or hot tools.

The Planrs Academy kits fix this by using safe, pre-wrapped magnetic clips that snap together easily. No sharp tools or hot parts are allowed!

| Step 1: Classroom Instructions

Tell the children to hold up the small, round, silver battery carefully between their fingers to check its sides before making any connections.

Step 1 Action Checklist:

1. Find the Plus Mark: Look closely at the flat, smooth side of your silver battery and find the tiny plus symbol (+).

2. Get the Red Wire: Pick up the bright red power wire clip from your box compartment.

3. Slide and Snap: Slide the shiny metal mouth of the clip onto the edge of the silver battery until you feel a fun, solid 'Click!'

SAY THIS TO THE KIDS:

"Look at that, my awesome little engineers! By snapping our clip on, we have opened up the main gate of our electrical water tank. The power is now safe and flowing down our red wire highway, waiting happily for our next command!"

8. How to Build: Step 2

Adding Our Gatekeeper Switch

If we connect our power wires straight from the battery into our lights without a switch, the lights will stay turned on all the time!

The battery would run out of energy in just a few minutes. Plus, our board game would lose its fun because there would be no distinct rolling action. Step 2 introduces our system gatekeeper.

Step 2: Classroom Instructions

Help the students attach the wire paths to the tiny pins on the push-button module.

Step 2 Action Checklist:

- 1. Find the Button:** Look for the tiny square block that has a bright, colorful round button on top.
- 2. Hold the Red Line:** Take the loose metal tip at the end of your positive red wire line.
- 3. Push into the Port:** Gently push that metal tip straight into the little opening on the left side of your button block.

 **Remember: No scissors, tape, or glue are needed at all!**

SAY THIS TO THE KIDS:

"We have just added a smart gatekeeper to our circuit! The power is running down the red highway, but it has now hit a locked gate at our button. It is waiting patiently for your thumb to press down and let it cross!"

9. How to Build: Step 3

Closing Our Magic Circle

The final step connects our button gatekeeper to the display screen, routing energy back into the power cell to establish our complete loop line. This step anchors our display system cleanly inside the primary protective housing frame.

Step 3: Classroom Instructions

Children will plug in the last few wires to light up the display screen during active testing.

Step 3 Action Checklist:

- 1. Connect Blue Wire:** Take the blue wire terminal coming out from the right side of your push-button gatekeeper.
- 2. Plug Into Screen:** Plug that blue terminal directly into the base socket of your central LED matrix block.
- 3. Return the Path:** Take the final black return wire coming from the bottom of the LED block, and clip it securely to the negative (–) face of your battery.
- 4. Fit into Shell:** Press the whole connected assembly gently down into the pre-made slots inside your colorful plastic housing shell.

The Grand Magic Test!

SHARE THIS EXCITING MOMENT:

"Now, let's test our hard work! Press down firmly on the button with your thumb. Wow! Look at that magic glow! The bright light dots are cycling rapidly and showing a surprise random number every single time you press down! Your temple circle is complete, and your custom electronic dice is ready for game time!"

10. Important Rules to Keep Kids Safe

Careful Design Limits for Young Classrooms

Teaching engineering to very young children requires careful safety steps to keep everyone physically safe and emotionally happy. Teachers and kit makers must follow these four basic rules:

Rule 1: No Sharp Tools & No Hot Solder

Kits and stories must never use scissors, knives, or hot soldering irons. Grade 1 students are still practicing their fine motor control. All wire connections must use pre-cut sleeves and simple click-fit magnetic leads.

Rule 2: No Main Wall Outlets

Children must never work near wall plug points or large power adapters. Systems are strictly limited to small, enclosed 3V DC coin-cell batteries. This provides absolutely zero risk of shock, ensuring total physical safety.

Rule 3: Use Local, Familiar Examples

Do not talk about foreign examples like adult casino dice patterns or big electronic roulette wheels. Keep examples focused on things kids know well, like local family board games or fun school playground activities.

Rule 4: Use Happy, Encouraging Words

Never use negative words like 'Circuit Failure', 'Short Circuit Defect', or 'Assembly Error'. If a child's light does not turn on, reframe it as a cute little sleeping firefly:

"Oh look, our little Jugnu (firefly) is resting! Let's check our round walking path together to wake it up!"

11. Tracking Our Success

How We Measure What Kids Learned

To see how well our lesson worked across all Planrs Academy classrooms, we track three easy goals during the session:

Goal A: Building Success

- **Target:** 100% of Students Finish
- Every single child should leave the class with a fully working, safely enclosed LED Dice that throws a surprise number whenever they press the button.

Goal B: Remembering the Concepts

- **Target:** 90% Correct on the Part Game
- During the wrap-up conversation, kids should easily match the parts to their real-world jobs: identifying the battery as the energy tank, the button as the gatekeeper, and the LED as the glowing output.

Goal C: Sharing the Joy

- **Target:** Active Group Sharing
- At least 95% of the children should happily show off their working dice to their desk partner or teacher during the final game, feeling proud that they are junior creators!

Congratulations, Teacher!

You have successfully guided your young engineers through their first hardware adventure.